

## **Homework 1**

**Due:** Monday, 10th June 2024 at 1:00 PM

**Instructions.** Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

**Note on Academic Dishonesty.** Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

**Grading.** Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

**Question 1.**

Use a direct proof to show that the sum of two even integers is even.

**Question 2.**

Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

**Question 3.**

Use a direct proof to show that the product of two odd numbers is odd.

**Question 4.**

Prove that if  $n$  is a perfect square, then  $n + 2$  is not a perfect square.

**Question 5.**

Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

**Question 6.**

Prove that if  $x$  is rational and  $x \neq 0$ , then  $1/x$  is rational.

**Question 7.**

Prove that if  $m$  and  $n$  are integers and  $mn$  is even, then  $m$  is even or  $n$  is even.



**Question 8.**

Prove that if  $n$  is an integer and  $3n + 2$  is even, then  $n$  is even using

a) a proof by contraposition.

b) a proof by contradiction.

**Question 9.**

Prove the proposition  $P(1)$ , where  $P(n)$  is the proposition “If  $n$  is a positive integer, then  $n^2 \geq n$ .”

What kind of proof did you use?

**Question 10.**

Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

**Question 11.**

Prove that if  $n$  is a positive integer, then  $n$  is even if and only if  $7n + 4$  is even.

**Question 12.**

Prove that  $m^2 = n^2$  if and only if  $m = n$  or  $m = -n$ .

**Question 13.**

Show that these three statements are equivalent, where  $a$  and  $b$  are real numbers: (i)  $a$  is less than  $b$ , (ii) the average of  $a$  and  $b$  is greater than  $a$ , and (iii) the average of  $a$  and  $b$  is less than  $b$ .

**Question 14.**

Show that these statements about the real number  $x$  are equivalent: (i)  $x$  is rational, (ii)  $x/2$  is rational, (iii)  $3x - 1$  is rational.

**Question 15.**

Show that the propositions  $p_1$ ,  $p_2$ ,  $p_3$ , and  $p_4$  can be shown to be equivalent by showing that  $p_1 \leftrightarrow p_4$ ,  $p_2 \leftrightarrow p_3$ , and  $p_1 \leftrightarrow p_3$ .